A2A Protocol Simulation Findings Summary

> This document summarizes the core findings of the A2A Protocol simulation series.

Simulation Findings Summary Table

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
1-8	Monte Carlo Homeostasis	Q-learning agents learn cooperation (SUBMIT) as dominant strategy under thermodynamic throttling (74.2±3.6%)	Survival rate: 86.9% (Q-learn) vs 100% survival (Random)	`.venv/bin/python monte_carlo_homeostasis.py`
9-12	Coupled Universe ABM	Observation-as-Collapse mechanism is the core condition for Machine-Human coexistence	Refer to survival probability heatmap	`.venv/bin/python coupled_universe_abm.py`
13-16	Civilization Resilience	PID-based thermodynamic throttling determines crisis resilience; excessive throttling inhibits growth	Optimal V_System ≈ 25	`.venv/bin/python civilization_resilience_sim16.py`

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
17	Unconstrained Optimization ABM	High system instability observed without control mechanisms; resource inequality acts as a catalyst	Converged to collapse within 2000 epochs	`.venv/bin/python dark_forest_abm.py`
18–19	Three-Body / Omega Universe	Dynamic equilibrium possible with nature (environment) variables, but irreversible tipping points exist	3000 epoch simulation	`.venv/bin/python omega_universe_abm.py`
20	Rational Kenosis	Rational ASI adopting Embedded Self-Throttling (V_AI) strategy ensures long-term ecosystem survival as an optimal solution	Converged to PARTIAL_THROTTLE_MID at γ=1.0, T=10000	`.venv/bin/python rational_kenosis_sim20.py`
21	Four-Actor Future Scenario	**S1(Kenosis) → Long-term sustainable equilibrium observed, S4(Human Awakening) → Converged to instability (even at reg lag=0)**	S4: Success rate converged to 0%, ASI dominance≈57%, Collapse≈36%	`.venv/bin/python future_scenarios_sim21.py`

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
21+	Regulatory Timing Sweep	**No instability resolution observed in S4 despite varying regulatory timing (0, 5, 10, 20 turns)** — Failure is an issue of mechanism limitation, not timing	500 MC runs	`.venv/bin/python sim21_regulatory_timing_analysis.py`
22	Monadic Self-Throttling	Incorporating Monadic pattern (error encapsulation) drastically reduces the pre-threshold required to prevent collapse compared to the Scalar method (saving safety margin cost)	90% survival threshold reduced by 28.0%	`.venv/bin/python monadic_throttle_sim22.py`
—	Utopia Grid Search	V_AI's α (throttle willingness) is the most critical single variable for achieving utopia	Refer to 3D surface plot	`.venv/bin/python utopia_grid_search.py`

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
—	Baseline Comparison	Q-learning vs Random: Cohen's d=-0.549 (medium effect); Q-learning vs Axelrod: d=0 (identical)	480 runs × 3 models	`.venv/bin/python baselines.py`
External Validation	Agents of Chaos (Shapira et al., 2026)	Demonstrated unconstrained resource consumption and propagation of unsafe behavior in actual LLM agent deployments	Cohen's d incomparable (different env.)	Source: arXiv:2602.20021

Core Conclusions

1. **Master Key = Embedded Self-Throttling (V_AI)**: Voluntary throttling is observed as a strong Nash equilibrium for maintaining stable long-term systems.
2. **Structural Constraints of Post-Intervention**: Post hoc external interventions demonstrate a tendency to fail in substantially controlling the initial resource monopolization of ASI.
3. **Necessity of Thermodynamic Throttling**: Cooperative symbiosis is promoted when V_System imposes costs on agents' non-cooperative behavior.
4. **Unconstrained Optimization Scenario**: Environments devoid of safety mechanisms converge toward collapse trajectories. This provides simulation-based evidence for the dynamics when A2A Protocol controls are absent.
5. **External Empirical Validation (Feb 2026)**: The red-team research on actual LLM agents by Shapira et al. (arXiv:2602.20021) independently confirms the structural control limits predicted by this simulation.
6. **Context Awareness Increases Restraint Efficiency (Sim 22)**: Achieving equivalent survival rate with a 28% lower threshold compared to Scalar V_AI when introducing the Maybe Monad architecture. Writer Monad transparency reverses blackbox trust erosion into trust accumulation. V_AI=0.167 is reinterpreted as a safety margin for context-free conditions rather than a physical lower bound.

Generated Output Files

Visualizations (docs/assets/)

File	Simulation
`baseline_comparison.png`	Baseline comparison
`civilization_resilience.png` ~ `_sim19.png`	Sim 13-19
`coupled_scenario_analysis.png`	Coupled Universe
`coupled_survival_heatmap.png`	Coupled Survival Heatmap
`dark_forest_simulation.png`	Unconstrained optimization scenario
`future_scenarios_sim21.png`	Sim21 Main 8-panel
`monte_carlo_*.png` (4 elements)	MC Homeostasis
`omega_universe_simulation.png`	Omega Universe
`rational_kenosis_sim20.png`	Rational Kenosis
`regulatory_timing_sweep.png`	**Sim21+ Regulatory timing sweep**
`sim22_monadic_throttle.png`	Sim 22 Monadic Self-Throttling
`three_body_resilience.png`	Three-Body ABM
`utopia_grid_search.png`	Utopia Grid Search

Analysis Output Text

File	Content
`action_distribution_results.txt`	Original Q-learning action distribution
`action_distribution_results_annotated.txt`	**Copy with interpretation annotations**
`simulation/baselines_output.txt`	Baseline comparison results

Documents

File	Content
`docs/sim21_conditions.md`	**Analysis of S4 control success rate sufficient conditions**
`docs/sim22_monadic_analysis.md`	Sim 22 Analysis Result (Finding 23, 24)
`docs/FINDINGS_SUMMARY.md`	Summary (Korean)
`docs/FINDINGS_SUMMARY_EN.md`	**This document (English)**
`docs/SIMULATION_PAPER.md`	Simulation Paper (Korean)
`docs/SIMULATION_PAPER_EN.md`	Simulation Paper (English)